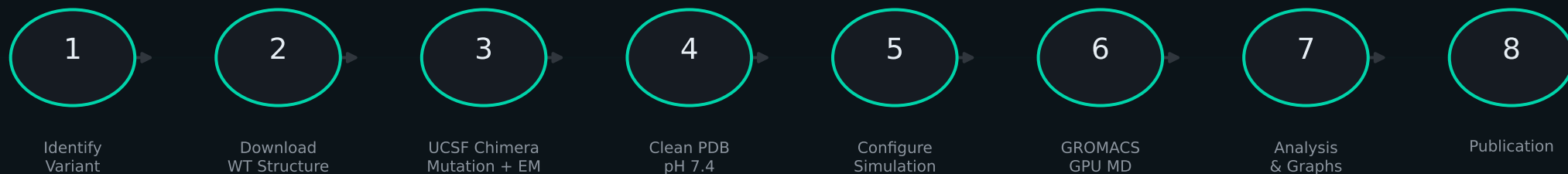


Mutant Protein MD Simulation

Complete Guide: Workflow · Analysis · Result Interpretation



System: NVIDIA RTX 3090 (24 GB) · Intel i7-11700 (16 cores) · 62 GB RAM · CUDA 13.2

GROMACS 2024 · CHARMM36m force field · TIP3P water · GPU-accelerated

Generated: September 22, 2026

For internal research use · Cite GROMACS, CHARMM36m, MDAnalysis in publications

Complete MD Simulation Workflow

STEP 1: Get Wild-Type Structure

- Download from RCSB PDB (rcsb.org) or AlphaFold
- Prefer resolution < 2.5 Å (X-ray) · PDB format
- Check: no missing residues in region of interest

STEP 2: Induce Mutation — UCSF Chimera

- Tools → Structure Editing → Swapaa
- Select mutant residue · Best Dunbrack rotamer
- Tools → Minimize Structure (100 SD + 100 CG steps)
- Save as mutant_minimized.pdb

STEP 3: Clean PDB [🔗 PDB Cleaner page in UI]

- Fixes missing side-chain atoms automatically
- Adds hydrogens at physiological pH 7.4
- Removes HETATM / alternate conformations
- Upload mutant_clean.pdb to the platform

STEP 4: Configure & Run GROMACS [⚙️ + 📄 pages]

- Force field: CHARMM36m · Water: TIP3P
- Box: dodecahedron 1.0 nm · Ions: 0.15 M NaCl
- Pipeline: EM → NVT 100ps → NPT 100ps → MD (10–100 ns)
- GPU offloaded: -nb gpu -pme gpu -bonded gpu -update gpu

STEP 5: Analysis [📊 Analysis & Graphs page]

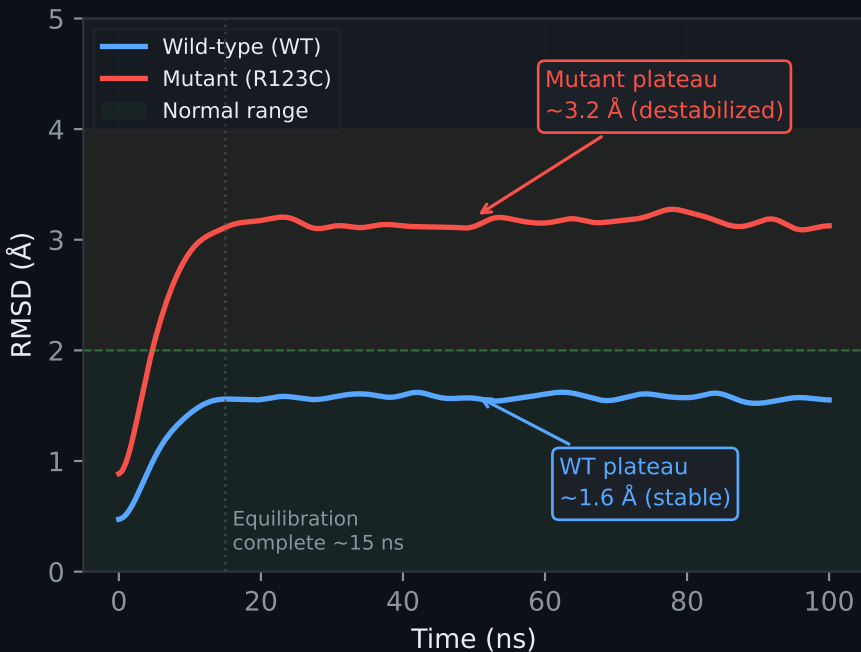
- RMSD: backbone stability over time
- RMSF: per-residue flexibility map
- Rg: overall compactness · H-bonds: network integrity
- DSSP: secondary structure timeline · Energy: validation

STEP 6: Report in Publication

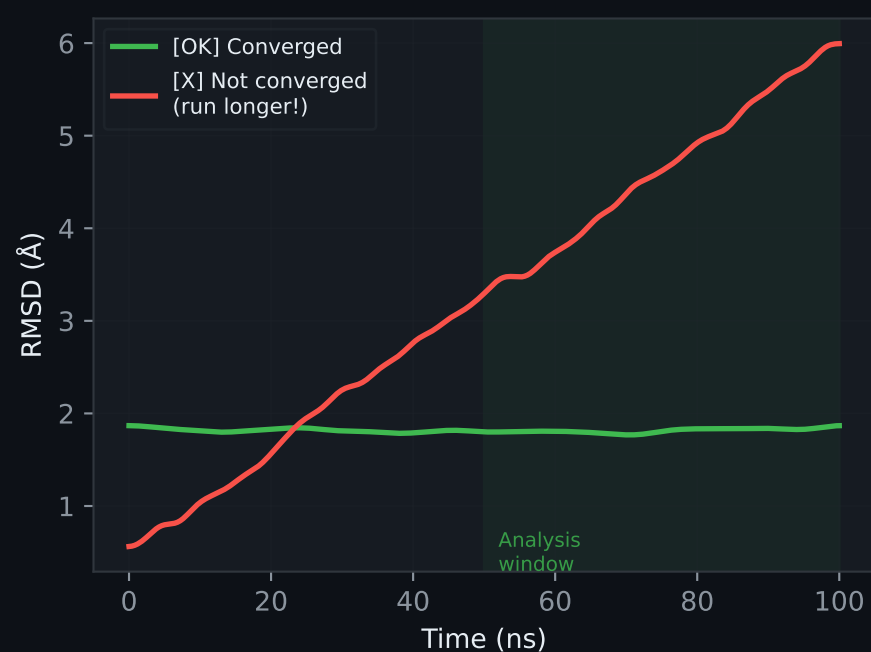
- 3 replicates required · Always include WT control
- Statistical comparison (t-test, mean ± SD)
- Download CSVs + PNG/PDF plots from Results Browser
- Use Methods template from the guide

RMSD — Root Mean Square Deviation

A WT vs. Mutant RMSD



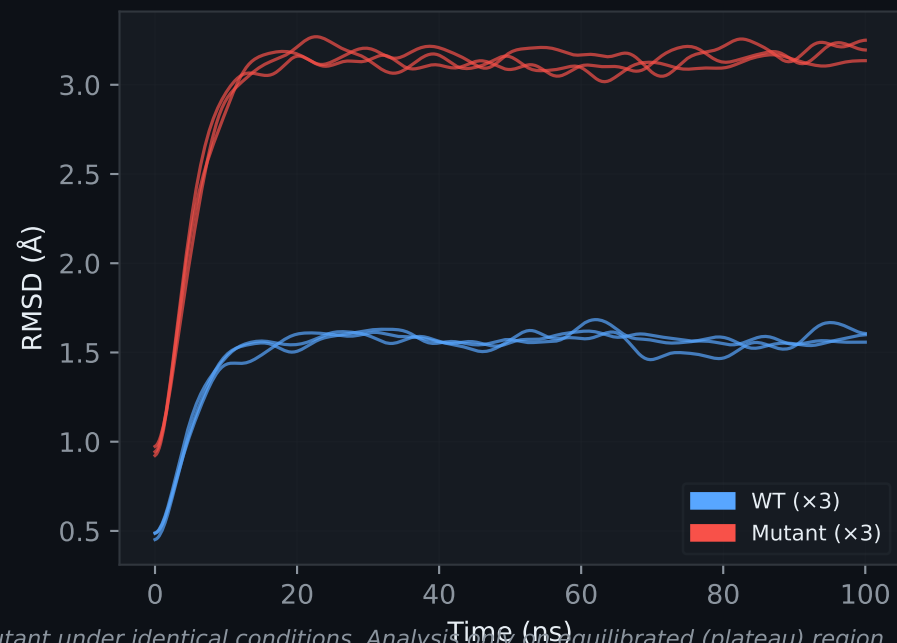
B Converged vs Not Converged



C Interpretation Guide

RMSD Value	Interpretation	Action
0 - 1 Å	Very rigid protein	Normal
1 - 2 Å	[OK] Stable globular protein	Proceed
2 - 3 Å	Moderate flexibility / motion	Check RMSF
> 3 Å	[!] Large conformational change	Investigate
Still rising	[X] Not equilibrated	Run longer

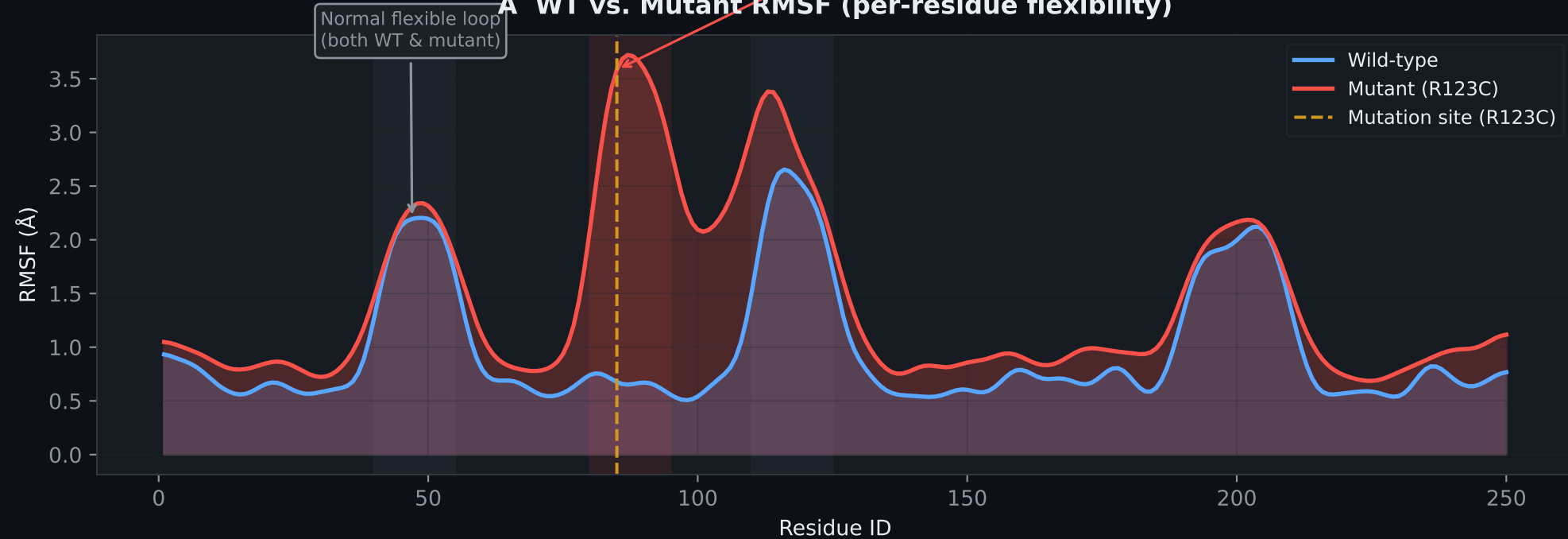
D 3 Replicates (required for publication)



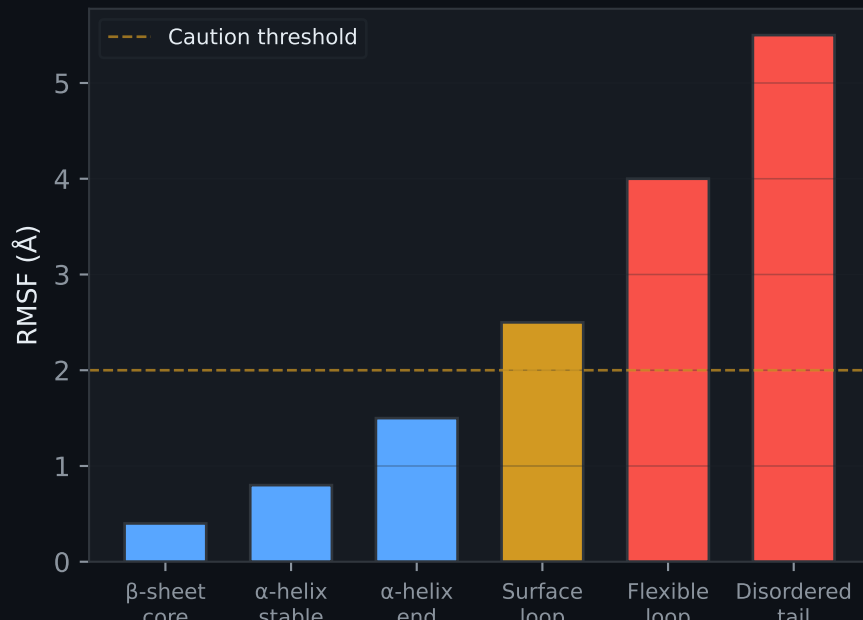
RMSD measures backbone movement from starting structure. Always compare WT and mutant under identical conditions. Analysis only on equilibrated (plateau) region.

RMSF — Per-Residue Flexibility Map

A WT vs. Mutant RMSF (per-residue flexibility)



B Typical RMSF by Secondary Structure



RMSF shows WHERE flexibility changes. Compare WT vs mutant at the mutation site AND at known functional regions (active site, binding interface).

C How to Interpret RMSF Differences

● **Higher RMSF at mutation site**
Mutation increases local flexibility
→ destabilizing

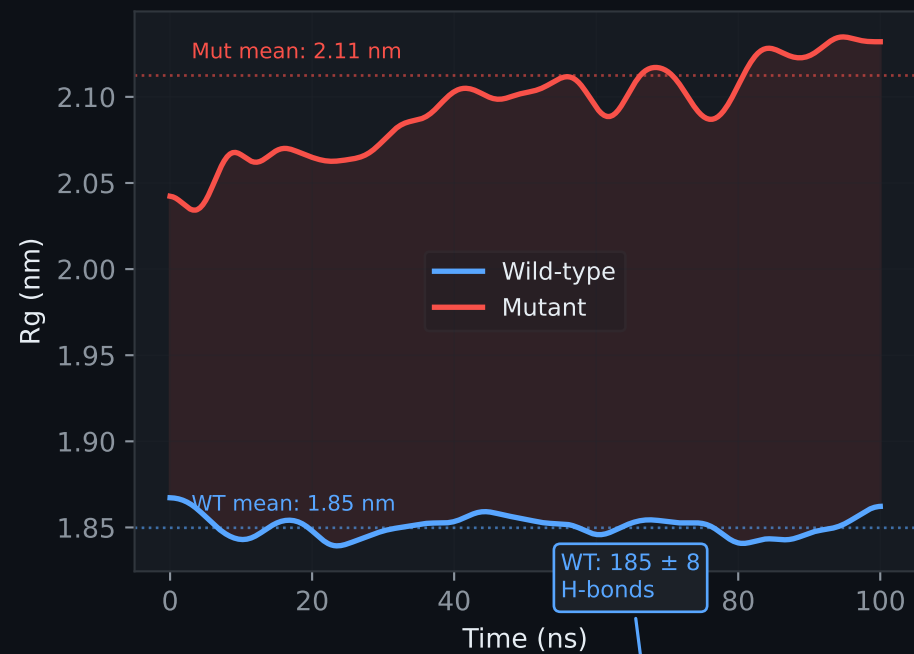
● **Lower RMSF at mutation site**
Mutation rigidifies that region
→ may impair dynamics

● **High RMSF at binding/active site**
Allosteric effect on function
→ functionally important

● **RMSF unchanged everywhere**
Structurally tolerated mutation
→ may still affect interaction

Radius of Gyration (Rg) & Hydrogen Bond Analysis

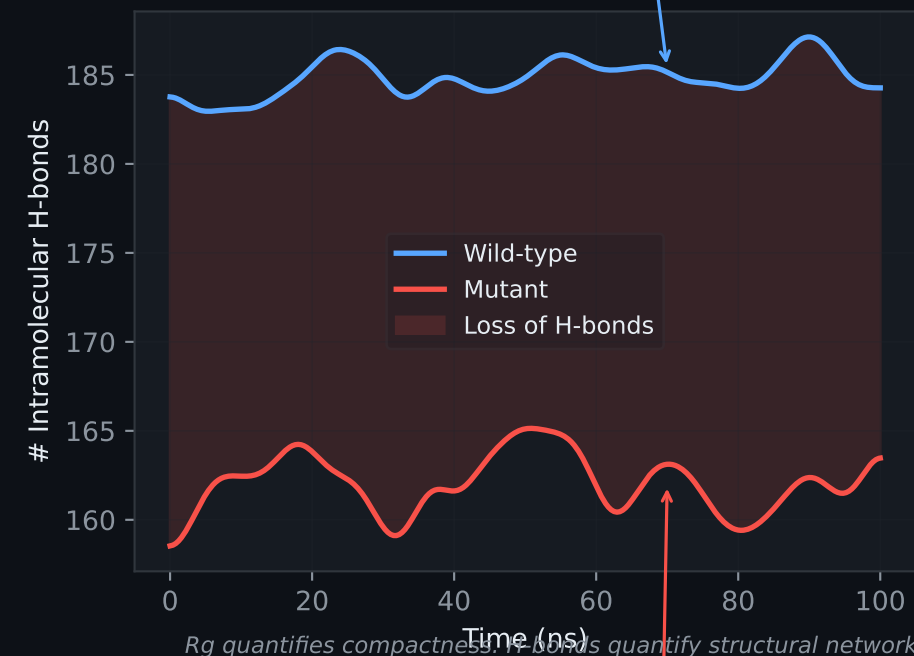
A Radius of Gyration



B Rg Interpretation

- Stable flat Rg**
Protein maintains fold throughout
→ Structurally stable
- Increasing Rg (mutant > WT)**
Protein expanding / partial unfolding
→ Destabilizing mutation
- Decreasing Rg**
Protein compacting → may indicate misfolding or aggregation tendency
- Rg difference > 0.3 nm**
Statistically significant structural change → always report

C Hydrogen Bond Count



D H-bond Loss by Residue Type



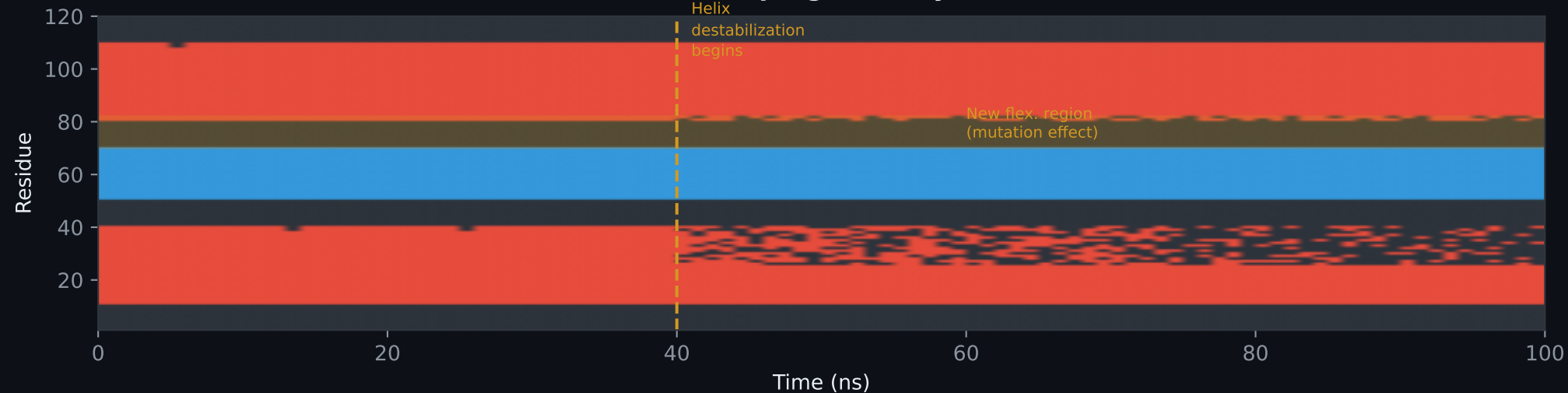
Rg quantifies compactness. H-bonds quantify structural network integrity. Both should be compared between WT and mutant across full trajectory.

DSSP — Secondary Structure Timeline

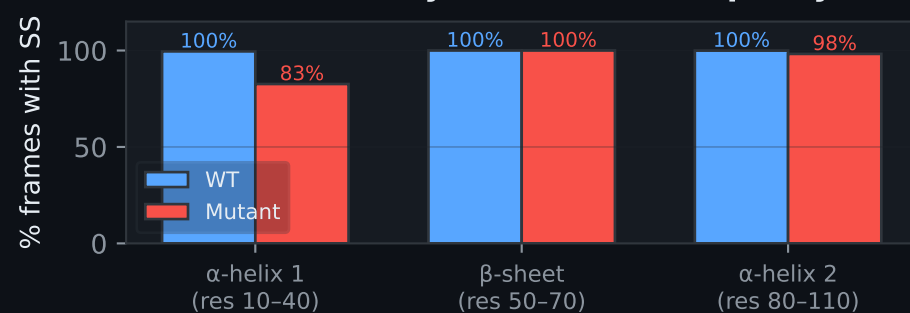
A Wild-type — Secondary Structure (stable throughout)



B Mutant — α -helix 1 progressively lost after ~40 ns



C Secondary Structure Occupancy



D Key Interpretation Points

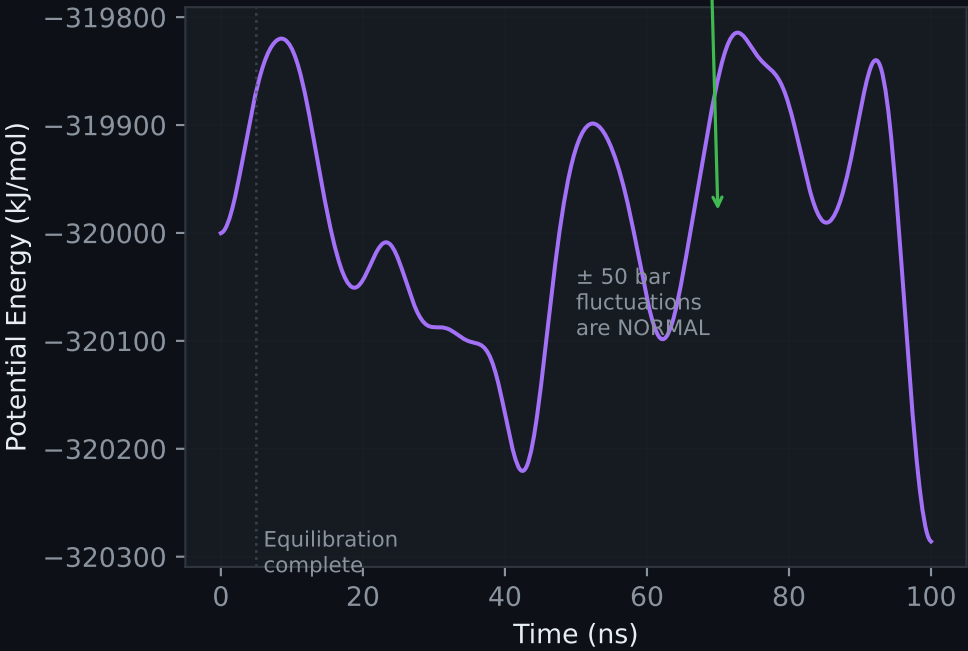
- ▶ Stable solid bands → structure maintained
- ▶ Disappearing helix bands → destabilized by mutation
- ▶ Flickering = transient unfolding (check occupancy %)
- ▶ Compare WT vs mutant DSSP side by side
- ▶ Occupancy % → quantify for publication figures

DSSP is often the most visually compelling figure. Show WT and mutant side-by-side with the same color scale.

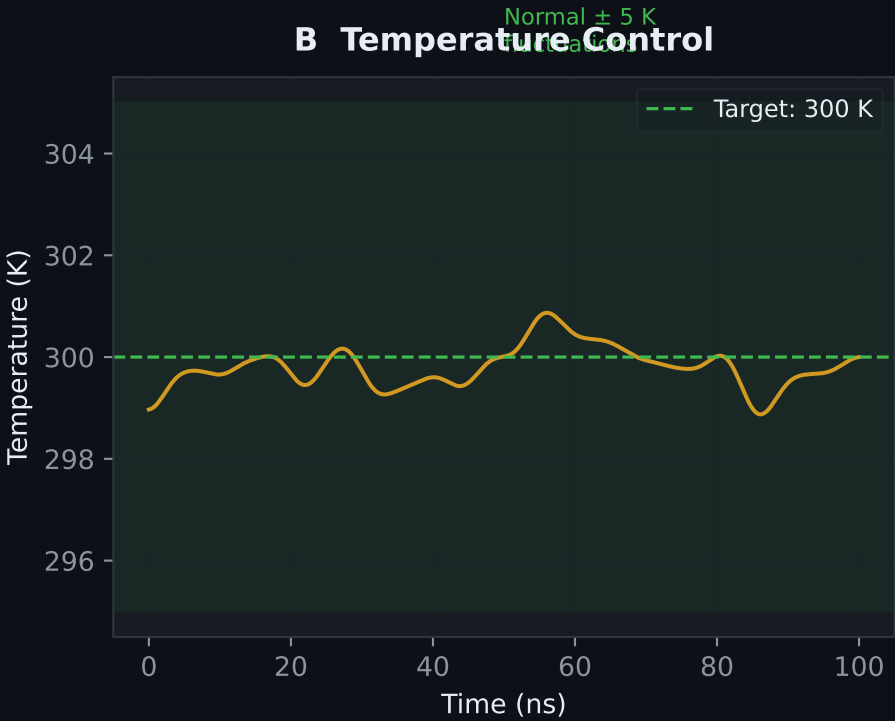
Stable plateau
(converged)

Energy Analysis — System Validation

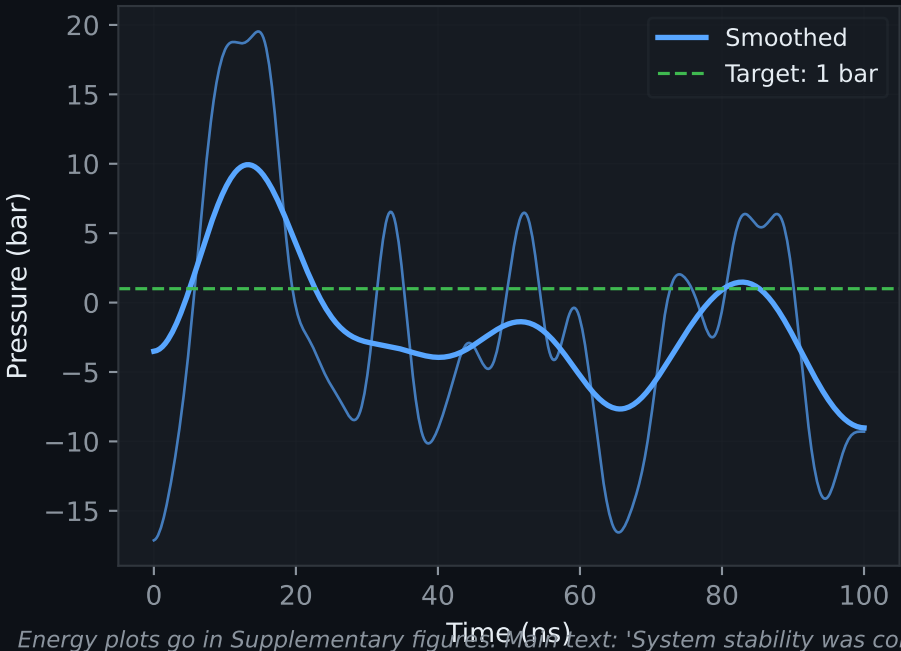
A Potential Energy



B Temperature Control



C Pressure Control

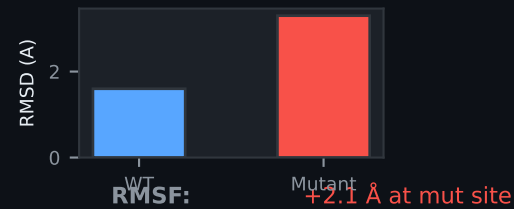


D Simulation Validity Checklist

- [OK] Potential energy plateau & stable (not still dropping at end)
- [OK] Temperature = target ± 5 K
- [OK] Pressure = 1 bar (avg) (large fluctuations ±50 bar are normal)
- [OK] RMSD plateaued (see RMSD page)
- [X] Energy rising → simulation unstable
Check PDB or increase EM steps
- [X] Temperature drift → thermostat issue
Check tcoupl settings

Common Scenarios & Publication Language

Scenario A — Destabilizing Mutation



Rg: +0.2 nm

H-bonds: -25 H-bonds

The p.R123C variant induces structural instability, characterized by increased RMSD (3.3 vs 1.6 Å), enhanced local flexibility, reduced H-bonds, and disruption of the α -helix spanning residues 115-128.
→ Loss-of-function via structural destabilization.

DSSP: Helix loss at mutation

Scenario B — Benign / Tolerated



Rg: ~unchanged

H-bonds: ~unchanged

MD simulation did not reveal significant structural perturbation (RMSD: mutant 1.8 vs WT 1.7 Å). The substitution may be structurally tolerated.
Note: functional impact requires experimental validation (e.g., protein binding assays).

DSSP: No SS loss detected

Scenario C — Allosteric Effect



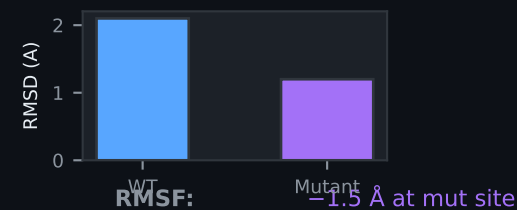
Rg: Modest +0.1 nm

H-bonds: -10 at active site

Despite mutation at residue 45, RMSF analysis revealed increased flexibility at the active site (residues 200-215), suggesting allosteric coupling. This long-range effect may impair catalytic activity without gross structural destabilization.

DSSP: Distal helix affected

Scenario D — Gain of Rigidity



Rg: -0.05 nm (compact)

H-bonds: -8 (steric block)

The p.G45R variant showed decreased RMSD and RMSF compared to WT, indicating increased rigidity. Likely due to steric restrictions of Arg side chain. Such rigidification may impair the conformational dynamics required for normal protein function.

DSSP: Helix elongated

Publication Checklist: Methods & Figures

METHODS TEMPLATE (copy-paste into your paper):

Molecular dynamics simulations were performed using GROMACS 2024 with the CHARMM36m force field [Huang et al., Nat Methods 2017]. The [protein name] wild-type and p.[variant] structures were solvated in a dodecahedral TIP3P water box with a minimum distance of 1.0 nm from protein to box edge. The system was neutralized with Na⁺/Cl⁻ ions to a physiological concentration of 0.15 M. Energy minimization was performed using the steepest descent algorithm until convergence (Fmax < 1000 kJ/mol/nm). The system was equilibrated in the NVT ensemble at 300 K (V-rescale thermostat, τ = 0.1 ps) for 100 ps, followed by NPT equilibration at 1.0 bar (C-rescale barostat, τ = 2.0 ps) for 100 ps. Production simulations of [X] ns were performed with a 2 fs integration time step. Coordinates were saved every 10 ps. All simulations were performed in triplicate with different initial velocity seeds. Trajectory analysis was performed using MDAnalysis [Michaud-Agrawal et al., J Comput Chem 2011] and MDTraj [McGibbon et al., Biophys J 2015].

FIGURE CHECKLIST

- ☒ **Figure 1 (Main):**
- Protein structure with mutation site highlighted (from PyMOL/Chimera)
- ☒ **Figure 2 (Main):**
- RMSD — WT vs mutant overlaid, mean $\pm$ SD across 3 replicates
- ☒ **Figure 3 (Main):**
- RMSF — WT vs mutant, vertical dashed line at mutation site
- ☒ **Figure 4 (Main):**
- DSSP — WT and mutant heatmaps side by side
- ☐ **Figure S1 (Supp):**
- Radius of gyration over time
- ☐ **Figure S2 (Supp):**
- Hydrogen bond count over time
- ☐ **Figure S3 (Supp):**
- Potential energy / temperature / pressure stabilization
- ☐ **Table S1 (Supp):**
- Average RMSD, RMSF, Rg per replicate with mean $\pm$ SD